Reg. No. | | | | | | | | | | |

[12EC833]

EIGHTH SEMESTER B.E UNIVERSITY EXAMINATION, JUNE - 2016

MULTIMEDIA COMMUNICATIONS

Time: 3 Hours

Maximum Marks:70

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. a. With a neat diagram, explain how voice mail and teleconferencing is supported in relation to speech only interpersonal communication involving both public (PSTN/ISDN) and private network. Also explain the role of voice mail server and audio bridge. **08**

   b. Explain the application QoS Parameters briefly. **06**

2. a. Derive the time to transmit the following digitized images at both 64kbps and 1.5 Mbps. **04**

   i. a 64X480X8 VGA-compatible image

   ii. a 1024X768X24 SVGA compatible image

   b. Explain Raster-Scan display architecture with a neat diagram. **06**

   c. Explain QCIF digitization format. **04**

3. a. Explain JPEG encoder with a neat diagram. **08**

   b A series of messages is to be transmitted between two computers over a PSTN. The messages comprise just the characters A through H. The Probability of each character is as follows: (probability indicates relative frequency of occurrence) A and B = 0.25, C and D =0.14, E,F,G and H =0.055 **06**

   i. use shannon's formula to derive the minimum average number of bits / character.

   ii. Use Huffman coding to derive the code word set and prove this is the minimum set by constructing Huffman code tree.

4. a. Explain third order Predictive DPCM signal encoder and decoder with a neat block diagram. **07**

   b. Explain MPEG-1 video bit stream structure and its format with a neat diagram. **07**

5. a. Explain MPEG-4 coding principles with the help of a neat diagram. **07**

   b. Explain Resource management in multimedia communications. **07**

6.	a.	Explain the descriptors of MPEG-7.	07
	b.	Discuss the salient features of JPEG 2000.	07
7.	a.	Discuss the mechanisms devised for combating transmission errors for real-time video transmissions.	07
	b.	Explain the process of layered video encoding.	07
8.	a.	Explain RTP packet format with a neat diagram.	08
	b.	Briefly explain RSVP protocol.	06